Manuel Eduardo Santiago Feria

Oaxaca, Mexico • 100% Remote / Local On-Site Only

Software Engineer | Backend & Full-Stack Developer (Node.js / NestJS / TypeScript / PostgreSQL)

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Portfolio: portfolio.nano-service.com

PROFESSIONAL SUMMARY

Computer Systems Engineering Graduate focused on Backend and Full-Stack Development. Experience building modular REST APIs and real-time web applications using **NestJS, Node.js, TypeScript, PostgreSQL, React, and Redis**. Focused on transactional integrity, concurrency conflict prevention, automated testing (**1,600+ unit & integration tests across projects**), and secure containerization with Docker.

TECHNICAL SKILLS

Backend & APIs: Node.js, NestJS, TypeScript, PHP (Laravel), Python, RESTful APIs, WebSockets (Socket.io), BullMQ, JWT / 2FA.

Databases, ORM & Storage: PostgreSQL (GIN Indexes / Full-Text Search), Redis (Cache & Queues), MySQL, TypeORM, Prisma ORM, MinIO / S3 Object Storage.

Testing, DevOps & Cloud: Jest (998+ Unit/E2E Tests), Pest PHP, Docker (Distroless / Compose), AWS, Google Cloud (GCP), Terraform, Nginx, Linux, Git.

Frontend & AI Integration: React 19, TanStack Query, Vue 3, Tailwind CSS, Vite, YOLOv8, ONNX Runtime DirectML, OpenCV.

PROFESSIONAL EXPERIENCE

Backend Developer — Professional Residency

2025 — 2026

Instituto Tecnológico de Oaxaca (TecNM / ITO) • IT Helpdesk & Asset Inventory System

`NestJS • TypeScript • PostgreSQL • Redis • TypeORM • Socket.io • MinIO S3 • React • Docker • Jest`

- **Modular Backend & Real-Time Architecture:** Developed a modular REST API in NestJS and real-time bidirectional communication via WebSockets (Socket.io) to manage IT support requests, hardware inventory, and incident tracking with Role-Based Access Control (RBAC).
- **Data Integrity & Concurrency Control:** Implemented explicit database transactions (TypeORM `QueryRunner`) and optimistic concurrency control (`@VersionColumn`) to prevent conflicts and data loss when multiple technicians update tickets and inventory simultaneously.
- **Fast Search, Caching & S3 Storage:** Optimized asset and ticket lookups using PostgreSQL Full-Text Search (`tsvector` and GIN indexes), paired with Redis in-memory caching, rate limiting, and S3-compatible object storage (MinIO) for ticket attachments.
- **Software Quality & Containerized Deployment:** Verified system stability through **998 automated tests across 143 test suites using Jest**, deploying the application in non-root Docker Distroless containers behind an Nginx reverse proxy.

KEY ENGINEERING PROJECTS

Core SaaS Platform — Backend & Web Architecture

2025 — 2026

Modular SaaS Platform with 2FA Authentication, Background Queues & Automated Testing (In Progress)

`NestJS • TypeScript • Prisma 7 • PostgreSQL • Redis • BullMQ • React 19 • ClamAV • Jest (627 tests)`

- **Security & Background Processing:** Designed a modular backend and web client featuring decoupled 2FA/TOTP authentication, RBAC permissions, asynchronous background job queues with BullMQ/Redis, S3 file storage with ClamAV antivirus scanning, and a test suite of **627 automated tests across 119 suites in Jest**.

AI Smart POS — Computer Vision Point of Sale System

2024 — 2025

Automated Product Detection, Multi-Object Tracking & GPU-Accelerated Inference

`Python • YOLOv8 • ONNX Runtime DirectML • ByteTrack • OpenCV • NumPy`

- **Real-Time Inference Pipeline:** Built an automated checkout system by training a custom YOLOv8 model achieving **97.1% mAP@50** on project validation datasets; exported to ONNX Runtime with DirectML GPU acceleration (~21 FPS) and integrated ByteTrack object tracking to prevent duplicate billing.

Laravel CMS & Reactive Blog

2024

Full-Stack Content Management System with Server-Side Reactivity & Automated Testing

`Laravel • PHP • Livewire Volt • MySQL • Spatie RBAC • Pest PHP (44 tests)`

- **Backend & Automated QA:** Developed a content management platform featuring server-driven reactivity (Livewire Volt), role/permission control (Spatie RBAC), CSV report exports, and **44 automated tests with 116 assertions using Pest PHP**.

EDUCATION

B.S. in Computer Systems Engineering

Instituto Tecnológico de Oaxaca (TecNM / ITO)

2021 — 2025 • Graduated (Professional Residency Concluded • Degree in Progress)

Languages: Spanish (Native) • English (B1/B2 — Technical Reading, Documentation & Professional Communication).

VERIFIED CERTIFICATIONS

AWS Solutions Architect, Cloud Practitioner & GCP Foundations

AWS (9af9ebae / 701bebc0) • Google Cloud (afaf62b7) • 2026

Cloud Security, Infrastructure, Data/ML & +10 Skill Badges

Google Cloud (14 Verified Badges on Credly: 88ab965e / 57991d00) • 2026